NS-760 (adravacetam): physicochemical and pharmacokinetic profile

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Background

NS-760 (adravacetam) is an investigational compound being studied in the Phase II BOREAS-2 study for the maintenance of cognitive performance during extended isolation and sleep restriction. This poster summarises its physicochemical properties and its pharmacokinetics in the rat.

Methods

Physicochemical properties were calculated from the structure and confirmed by measurement where indicated. Oral bioavailability and brain exposure were determined in male Sprague-Dawley rats after intravenous (1 mg/kg) and oral (5 mg/kg) dosing. Unbound brain-to-plasma ratio (Kp,uu) was derived from equilibrium dialysis of brain homogenate and plasma. Efflux was measured in MDCK-MDR1 monolayers.

Conclusions

NS-760 has high oral bioavailability and unrestricted brain exposure in the rat. No active transport by P-glycoprotein was identified. The data support the oral dosing used in BOREAS-2.

Physicochemical and pharmacokinetic data

Property	Value
Molecular formula	C51H80N10O17
Molar mass	1105.3 g/mol
Hydrogen-bond donors	9
Hydrogen-bond acceptors	23
Topological polar surface area	352 Å²
cLogP	-2.1
Oral bioavailability (rat)	74%
Brain/plasma ratio (Kp)	1.3
Unbound brain/plasma ratio (Kp,uu)	0.92
P-gp efflux ratio (MDCK-MDR1)	0.9; no active transport identified